IncidentDesk

Incident Timeline and Evidence Collection for Small Teams

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College: Vishwakarma Institute of Technology
Department: Computer Engineering

TEAM

Team Member 1: Ayush Shailesh Jadhav | GR Number: 1251070282 |
Email: ayush.1251070282@vit.edu

Team Member 2: Soham Shirish Jadhav | GR Number: 1251070826 |
Email: soham.1251070826@vit.edu

Guided By: Dr. Kaushalya Thopate, Department of Computer Engineering

Abstract

IncidentDesk is a proposed software-based tool that helps small IT teams carry out preliminary security-incident investigation. Users upload supported CSV security logs; the tool extracts events, normalizes timestamps, arranges events chronologically, applies basic rules to flag potentially suspicious patterns, calculates a SHA-256 hash of the evidence file, and generates a structured PDF investigation report. It is expected to make early investigation more organized and manageable.

Problem Description

Security events can be scattered across multiple log files, making it difficult for small IT teams to reconstruct incidents, connect events and identify suspicious sequences.

- Security records are scattered across multiple files.
- Investigators must manually compare timestamps.
- Suspicious sequences may be overlooked.
- Small teams may lack dedicated security-analysis tools.

Background

Authentication and other system events generate security records that can help investigators understand what happened [3].
Login successes, failed attempts, logouts and other available events each describe part of an incident.
Placing events in chronological order makes relationships between them easier to examine.
A small, accessible software tool can help organize these records for teams without dedicated security-analysis tools.
IncidentDesk applies this idea: it brings supported logs into one interface, orders events by time, and highlights sequences worth investigating.

Illustrative sample data

Time	Event Description
09:00	Successful Login - User A logged in
09:15	Failed Login - User B entered an incorrect password
09:16	Failed Login - Another login attempt failed
09:17	Successful Login - User B logged in successfully
09:45	Logout - User B logged out

Repeated failed logins followed by a successful login can be flagged for investigation. This pattern alone does not establish that an attack occurred.

Literature & RFC Review

1 Incident handling: NIST SP 800-61 Rev. 3 [1] treats incident response as part of cybersecurity risk management and stresses detection and analysis; IncidentDesk supports the early analysis step by organizing event records.

2 Forensics in incident response: NIST SP 800-86 [2] describes using forensic techniques, including combining multiple data sources to better understand an event, to reconstruct security incidents; IncidentDesk consolidates multiple log files into one timeline.

3 Log management: NIST SP 800-92 [3] covers generating, storing and analyzing computer security logs; IncidentDesk analyzes exported CSV logs and does not replace log collection infrastructure.

4 Evidence handling: RFC 3227 [4] gives guidelines for evidence collection and archiving; IncidentDesk records a SHA-256 hash of the uploaded evidence file so later changes can be detected, and works on uploaded copies.

5 Event data: RFC 5424 [5] defines the syslog message format including a TIMESTAMP field, showing why consistent timestamps matter; IncidentDesk normalizes timestamps from supported CSV logs and does not implement syslog.

6 Integrity hashing: FIPS 180-4 [6] specifies the SHA-256 algorithm used for the evidence hash (Python hashlib).

These references describe established practice; IncidentDesk is a small prototype and does not claim compliance with them.

Proposed Solution

IncidentDesk is a software-based platform for organizing supported security logs and assisting preliminary incident investigation.

1. Upload supported CSV log files → **2. Extract event details and timestamps** → **3. Normalize timestamps and arrange events chronologically** → **4. Apply basic rules to flag potentially suspicious patterns** → **5. Calculate a SHA-256 hash for the evidence file** → **6. Display findings and generate a PDF report**

Login and Logout Records

organizes available authentication events from supported logs

Failed Login Detection

flags repeated failed login attempts for further investigation

Incident Timeline

displays security events in chronological order

Suspicious-Pattern Identification

highlights potentially unusual sequences using simple rules

Evidence Integrity Verification

uses SHA-256 hashes to help detect changes to evidence files

Investigation Report Generation

generates a PDF containing the timeline, findings and evidence references

Important: Suspicious patterns require human investigation and are not proof of an attack. IncidentDesk is a proposed solution; features are planned for the initial prototype.

Solution Optimization

Optimization Proposed

Expected improvements over the current manual process (qualitative; no performance has been measured).

Current / Manual Process	Proposed IncidentDesk Approach
Scattered log files	Consolidated processing of supported log files
Manual timestamp comparison	Automatic chronological event sorting
Difficult-to-notice login sequences	Rule-based flags for repeated failed logins and other configured patterns
Manual evidence-integrity checks	SHA-256 hash generation and re-verification
Manually prepared investigation summaries	Structured PDF investigation reports
Limited visibility into event sequences	A unified incident timeline

Solution Architecture & Design

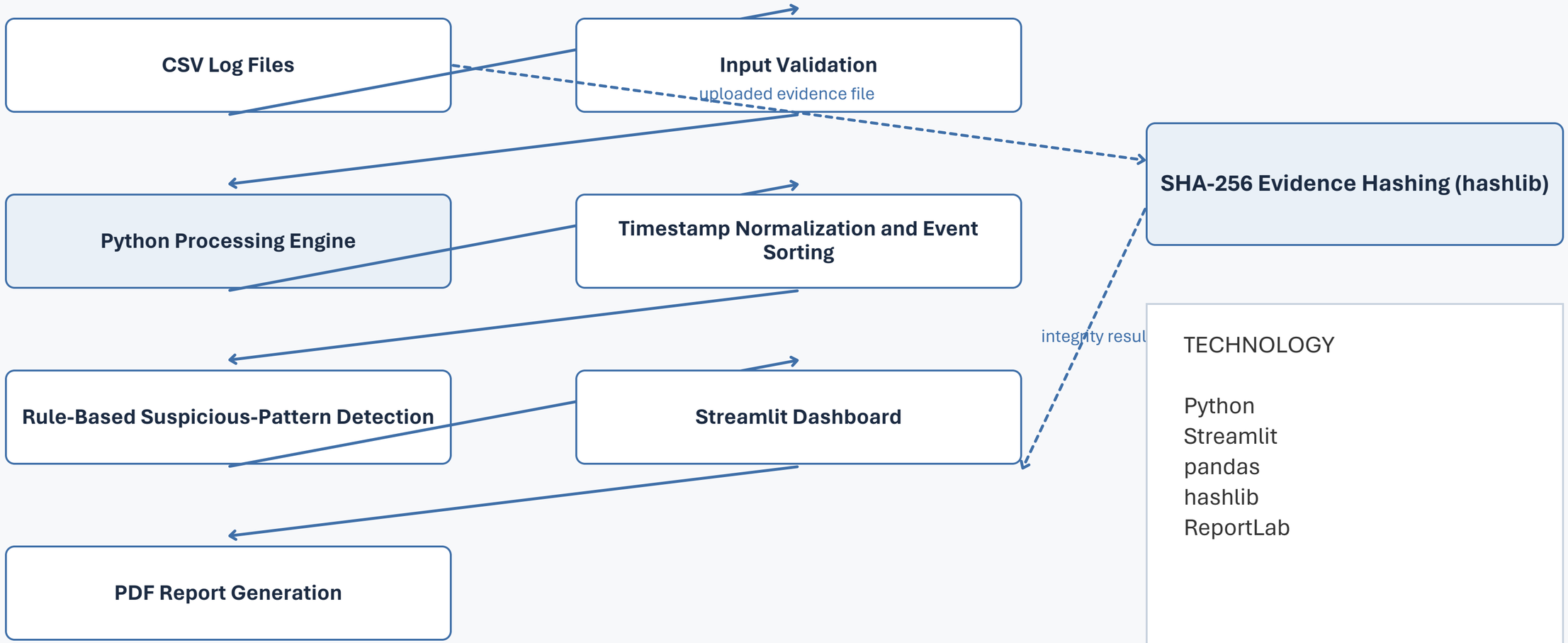


Figure 1: System Architecture UML Diagram

Data Flow Diagram

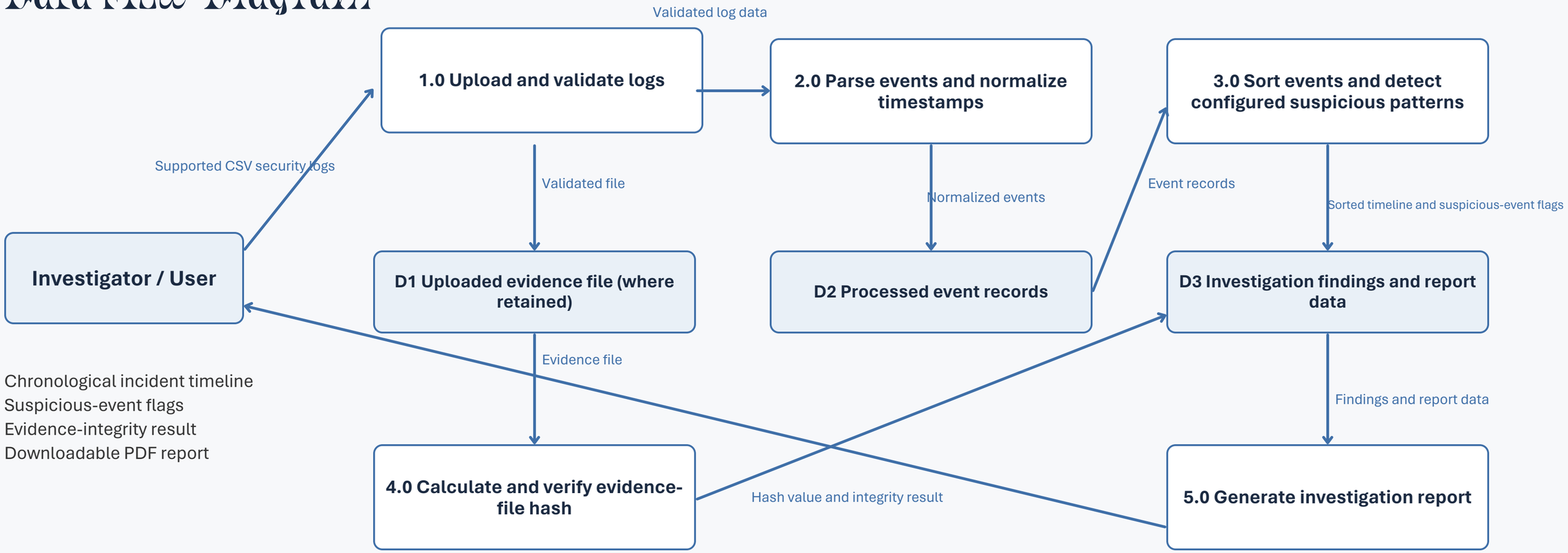


Figure 2: Data Flow Diagram (DFD)

Deployment Timeline

DAY 1

Build the upload interface

Prepare sample security logs

Implement timestamp processing and timeline generation

Add basic suspicious-pattern detection

DAY 2

Implement SHA-256 hashing

Generate PDF reports

Integrate all components

Test the workflow and prepare the demonstration

These are planned milestones, not completed work. The first prototype uses uploaded sample logs and does not claim to perform real-time monitoring.

Bibliography

[1] National Institute of Standards and Technology, "Incident Response Recommendations and Considerations for Cybersecurity Risk Management: A CSF 2.0 Community Profile," NIST SP 800-61 Rev. 3, Apr. 2025. doi: 10.6028/NIST.SP.800-61r3. https://csrc.nist.gov/pubs/sp/800/61/r3/final

[2] K. Kent, S. Chevalier, T. Grance and H. Dang, "Guide to Integrating Forensic Techniques into Incident Response," NIST SP 800-86, Aug. 2006. doi: 10.6028/NIST.SP.800-86.

[3] K. Kent and M. Souppaya, "Guide to Computer Security Log Management," NIST SP 800-92, Sep. 2006. doi: 10.6028/NIST.SP.800-92.

[4] D. Brezinski and T. Killalea, "Guidelines for Evidence Collection and Archiving," IETF RFC 3227 (BCP 55), Feb. 2002. https://www.rfc-editor.org/info/rfc3227

[5] R. Gerhards, "The Syslog Protocol," IETF RFC 5424, Mar. 2009. https://www.rfc-editor.org/info/rfc5424

[6] National Institute of Standards and Technology, "Secure Hash Standard (SHS)," FIPS PUB 180-4, Aug. 2015. doi: 10.6028/NIST.FIPS.180-4.

Conclusion

IncidentDesk aims to make preliminary cybersecurity investigation more manageable for small teams by organizing supported security logs, reconstructing event timelines, highlighting potentially suspicious sequences, verifying evidence integrity and producing structured reports.

Centralized organization of security evidence
Clearer reconstruction of event sequences
Easier identification of patterns requiring investigation
Practical software-only implementation for small teams

These are expected benefits; actual performance has not yet been measured.

Thank You